A Century of Change: Visualizing the Ecological and Physical Transformation of Chicago

Andrew McGallian
The University of Chicago
Environmental Science Master's Student 2025–2026

1 Introduction

Chicago's physical and ecological environment is a dynamic and layered system—shaped by centuries of natural processes, infrastructural decisions, and environmental policy. Over the last 150 years, the city has undergone dramatic transformations such as the reversal of the Chicago River, the expansion and erosion of its Lake Michigan shoreline, the removal and replanting of tree canopy, and the implementation of green infrastructure to mitigate urban heat and stormwater impacts. These changes are imprinted in the city's landscape, but they remain largely unexamined as an integrated whole.

While many studies have investigated individual indicators—urban heat, vegetation loss, shoreline retreat, or air pollution exposure—these analyses tend to be siloed. Most focus on narrow timeframes, a single environmental factor, or isolated case studies. Public platforms, when they exist, typically visualize static slices of environmental data or limited geographies. There is currently no accessible, spatial-temporal framework that brings together these layers of ecological change in Chicago's built and natural environments to tell a broader story.

This project proposes to fill that gap. By integrating geospatial analysis, archival cartography, remote sensing, and predictive modeling, I will develop a dynamic, exploratory platform that visualizes how Chicago's environment has evolved across multiple ecological dimensions. From shoreline position to vegetation recovery, from thermal fingerprints to green infrastructure efficacy, the platform will provide a citywide, view of environmental change over time. The goal is not just to inform academic research, but to support public understanding, environmental justice, and future planning.

Research Question

How has Chicago's physical and ecological environment changed over the last 150 years, and what spatio-temporal patterns and policy impacts emerge when synthesizing historical and contemporary data on shoreline evolution, watershed dynamics, vegetation, urban heat, pollution, biodiversity, and green infrastructure interventions?

2 Methods and Data

This project will use a combination of geospatial analysis, remote sensing, archival mapping, and predictive modeling to create an interactive spatial visualization of Chicago's environmental transformation.

2.1 Primary Data Sources

- **Shoreline/Watershed Dynamics:** USGS historical topographic maps, digitized city planning documents; shoreline extents and watershed boundaries will be digitized and analyzed over time using archival and geospatial datasets.
- **Urban Heat:** Satellite data such as Landsat, Sentinel, GOES, MODIS. Landsat specifically can be accessed via EarthExplorer, used to calculate historical land surface temperature (LST) and thermal patterns across the city.
- **Vegetation and Urban Canopy:** Normalized difference vegetation index (NDVI) data derived from satellite imagery; high-resolution vegetation and canopy data from HiTAB-Chicago and The Morton Arboretum, which offers tree and building height classification and could be helpful with model training to gather data where sparse or unavailable.
- **Pollution and Air Quality:** Environmental Protection Agency (EPA) datasets and available municipal datasets.
- **Soil/Toxins:** Community maps and soil sampling such as data sourced from this Chicago Soil Contamination StoryMap and relevant city or federal environmental reports.
- **Biodiversity/Wetlands:** Archived environmental assessments, ecological surveys, and wetland inventories from Chicago city or reports, surveys from the Chicago Wilderness Alliance, and historical records.
- **Green Infrastructure:** Implementation records and geospatial layers from the Chicago Data Portal, including green roof locations, tree planting initiatives, permeable alley programs, and stormwater infrastructure.

It is notable that these are datasets and sources I have been able to find from a preliminary examination, this does not include all primary data sources that could be of use.

2.2 Analytical Tools and Platforms

- ArcGIS and QGIS for spatial analysis and map integration
- Python libraries such as GeoPandas, Folium, and Rasterio for data wrangling and visualization
- Google Earth Engine for remote sensing data processing

- Machine learning techniques (e.g., CNNs, random forest) to extract features from historical maps and model future environmental trends

3 Project Plan and Deliverables

3.1 Timeline

- **Fall 2025:** Data collection and preprocessing—digitize and georeference historical maps, compile satellite imagery and environmental datasets
- **Winter 2026:** Conduct spatial analysis, develop statistics and possibly composite indicators, begin development of interactive dashboard
- **Spring 2026:** Finalize interactive visualization, document workflows, and create instructional materials (e.g., reproducible notebooks, workshop module)

3.2 Deliverables

- An interactive map-based dashboard that visualizes environmental change across multiple indicators over the past, present, and future.
- A documented methodological pipeline
- Educational resources suitable for classroom or workshop use

4 Relevance and Qualifications

This project builds directly on my academic background in Environmental Science and GIScience, as well as my forthcoming coursework in Environmental Data Science at the University of Chicago. I have a foundation in both the scientific principles underlying urban environmental change and the spatial tools required to investigate them. These include geospatial analysis, remote sensing, and data visualization, which I have applied in multiple research and teaching contexts.

Through the Environmental Frontiers initiative, I helped lead a team to design and deploy mobile microclimate sensors across the Chicago to collect fine-scale temperature and humidity data. We integrated this with satellite-derived metrics—LST and NDVI—to spatially examine urban heat exposure. I presented this research at the American Geophysical Union 2024 Meeting, where our project contributed to broader conversations around climate resilience, data justice, and hyperlocal sensing.

As a MAPSCorps Research Instructor for that same project, I mentored a cohort of high school students as they explored research and these same environmental questions. I guided them in using tools such as ArcGIS Online, GeoPandas, and Matplotlib to map temperature distribution, analyze census-based vulnerability indices, and develop place-based data stories. Their final deliverables- an ArcGIS StoryMaps and poster presentation—were featured at a citywide research symposium.

That experience deepened my appreciation for the public-facing power of geospatial analysis and storytelling, and affirmed my interest in developing tools that are accessible, instructive, and impactful.

Together, these experiences demonstrate not only my technical capabilities in spatial and environmental analysis, but also a deep commitment to educational outreach and interdisciplinary engagement. I am well prepared to take on an independent visualization project that is both methodologically rigorous and widely usable.

5 Broader Impact

Aligned with some of the goals of the Data Visualization Fellowship, this project is designed to serve as both a tool for research and a public-facing resource. By synthesizing historical and contemporary data through spatial visualization, it provides a novel way to engage with and understand long-term patterns of environmental change in urban settings. In doing so, it supports academic inquiry, civic awareness, and evidence-based policy discussion around topics like green infrastructure, climate resilience, and environmental justice.

The final product will be more than a dashboard—it will be a dynamic, interactive platform that enables students, researchers, planners, and community members to explore the layered ecological history of Chicago. The platform's modular structure and open documentation will make it adaptable for use in classroom settings, university classes, community data workshops, and municipal planning offices.

This fellowship also offers a unique opportunity for me to engage with library archives and digital scholarship infrastructure. Through this collaboration, I aim to develop workflows and instructional materials that demonstrate how archival cartography, remote sensing, and computational tools can be combined to support place-based environmental storytelling.